

INFORMATION SECURITY OVERVIEW

Information security refers to the practice of protecting information and information systems from unauthorized access, use, disclosure, disruption, modification, or destruction. The main objective of information security is to ensure the confidentiality, integrity, and availability of data. These three principles are commonly known as the CIA triad.

Confidentiality means that information is accessible only to those who are authorized to have access. Organizations implement access control mechanisms such as authentication, encryption, and permissions to protect confidential information.

Integrity ensures that data remains accurate and unaltered unless modified by authorized users. Techniques such as hashing, digital signatures, and checksums are used to verify that data has not been tampered with. Availability ensures that systems and data are accessible when needed. Organizations maintain availability through backups, redundancy, disaster recovery plans, and reliable infrastructure. There are many types of threats to information security. Malware, including viruses, worms, and ransomware, can compromise systems and steal or destroy data. Phishing attacks attempt to trick users into revealing sensitive information such as passwords or credit card numbers. Another common threat is unauthorized access. Attackers may attempt to break into systems using stolen credentials, brute-force attacks, or exploiting vulnerabilities in software.

To protect against these threats, organizations implement multiple security measures. These include firewalls, intrusion detection systems, antivirus software, encryption, and security policies. Regular security training for employees is also important because human error is one of the most common causes of security breaches. Risk management is another important part of information security. Organizations must identify potential threats, assess vulnerabilities, and implement controls to reduce risk to an acceptable level.

In addition, security standards and frameworks such as ISO 27001 and the NIST Cybersecurity Framework provide guidelines for implementing effective information security management systems. In conclusion, information security is essential for protecting sensitive information, maintaining trust, and ensuring the reliable operation of information systems. As technology continues to evolve, organizations must continuously update their security strategies to address new and emerging threats.